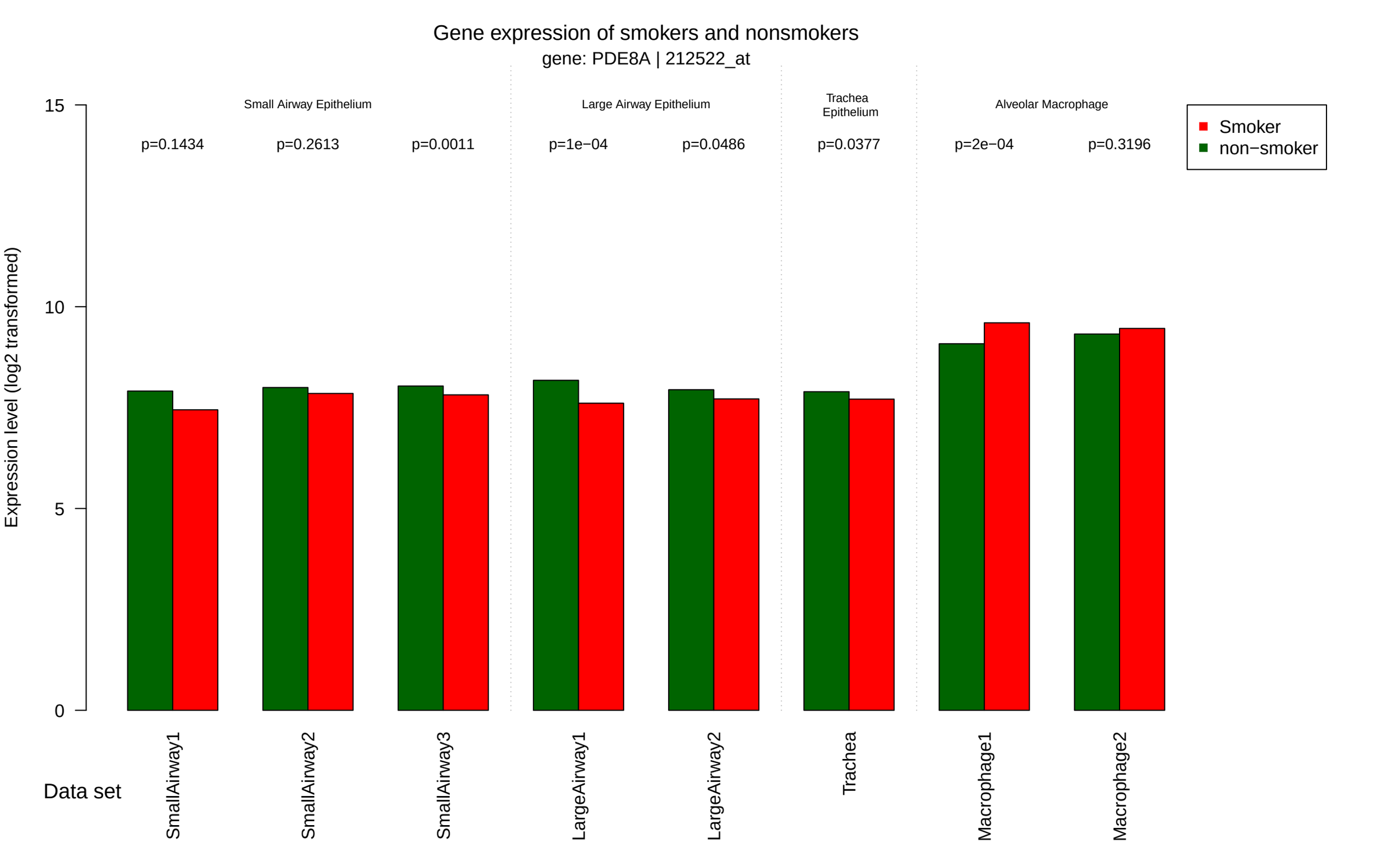
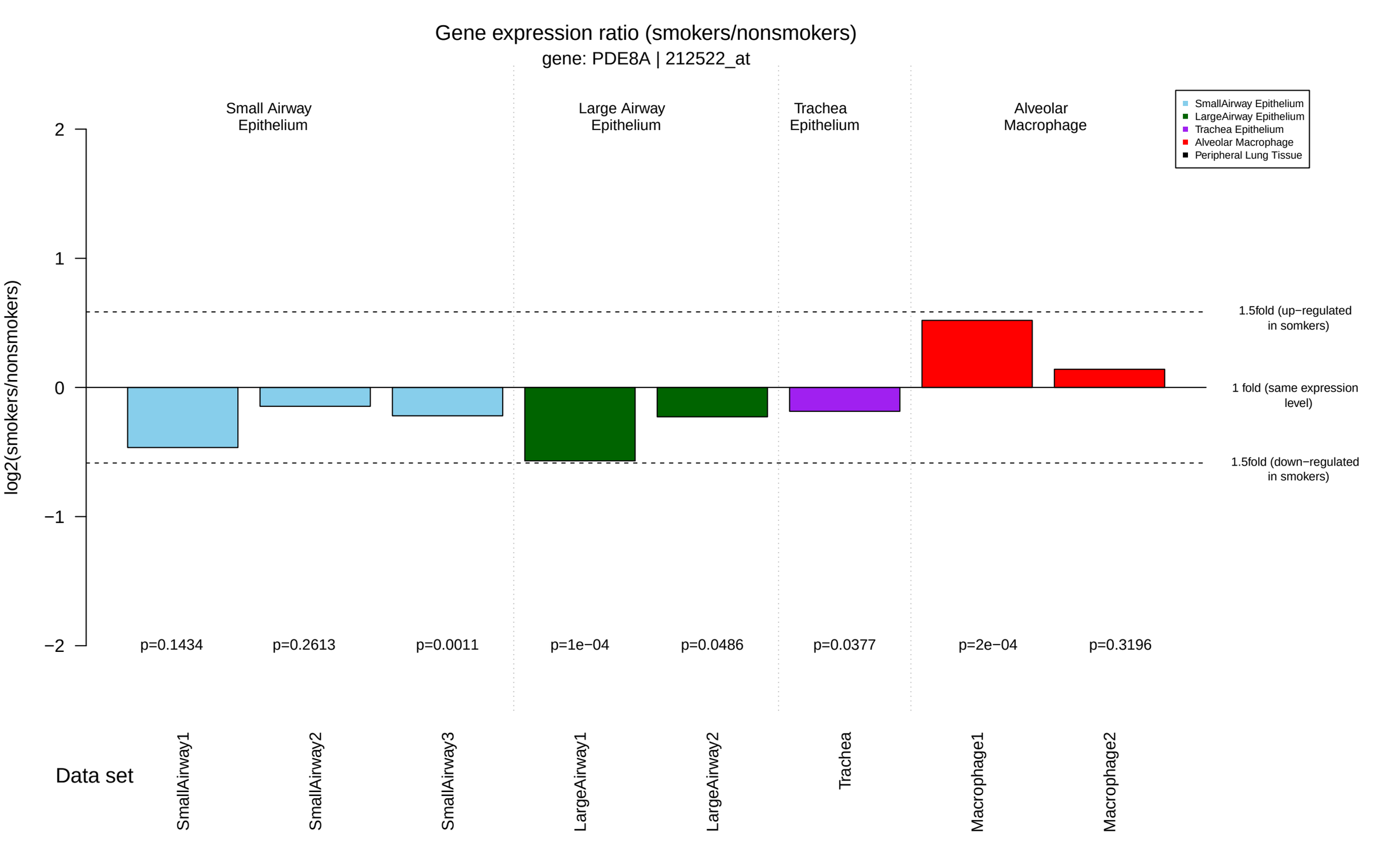
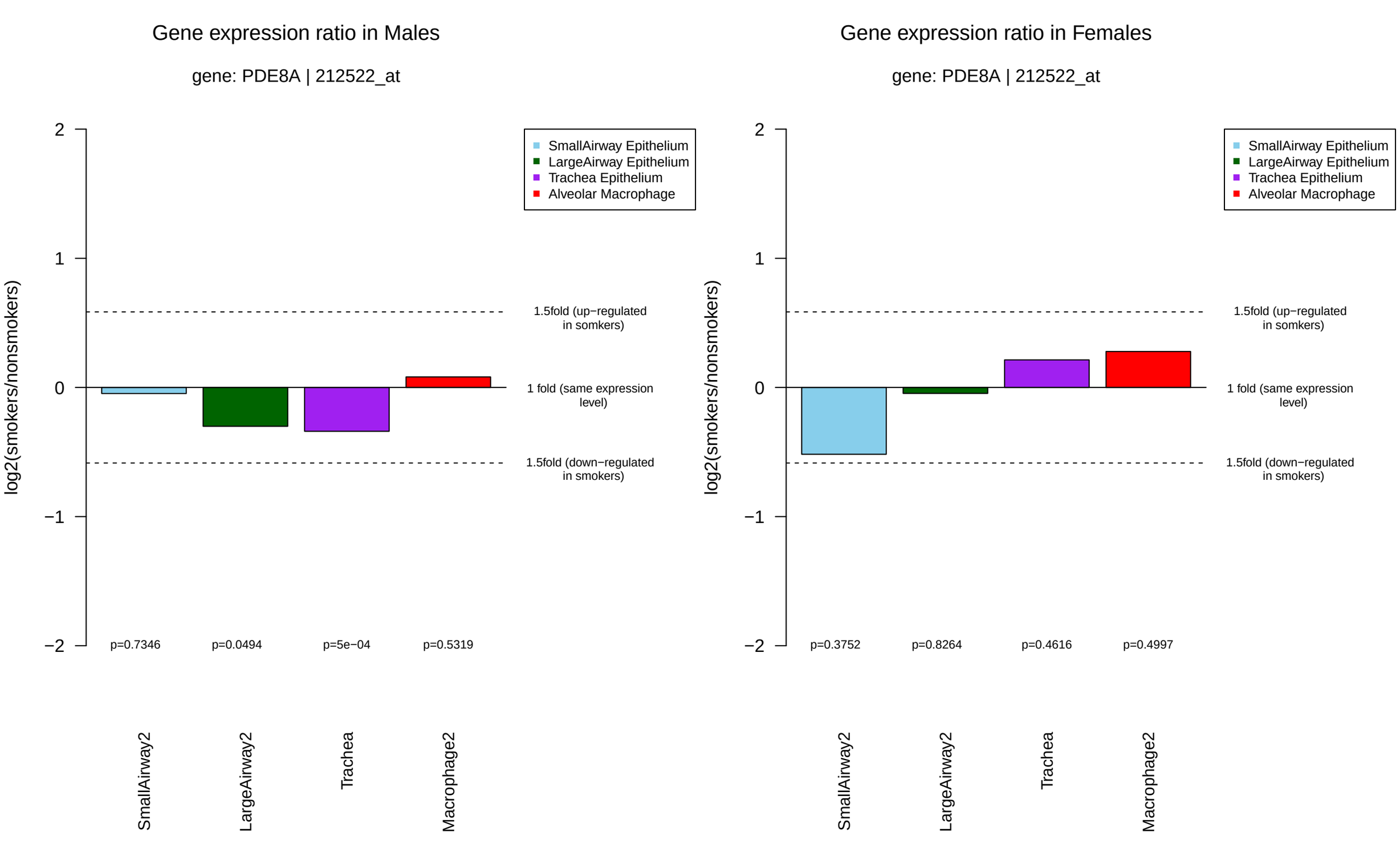
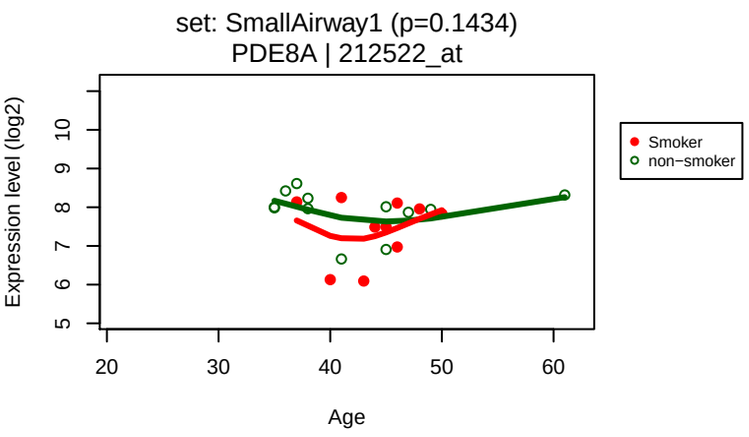
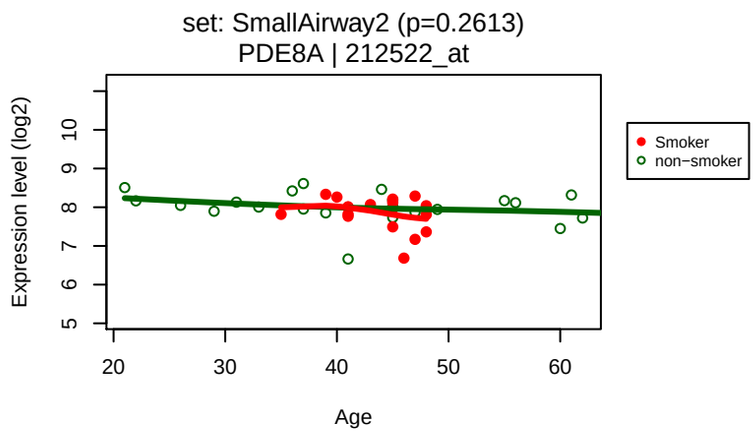
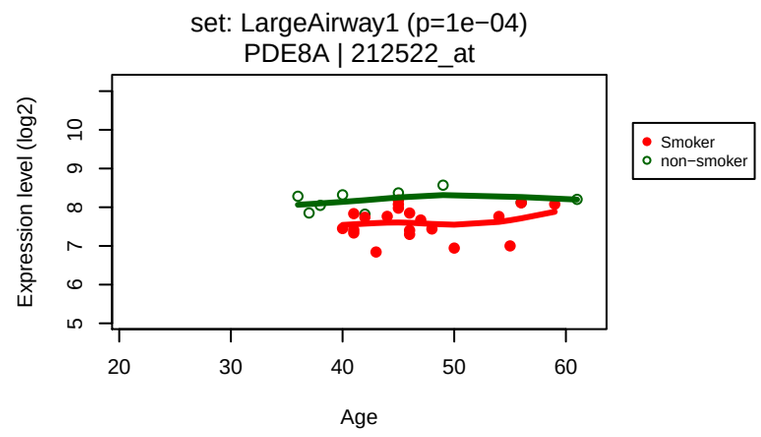

Gene expression of smokers and nonsmokers
gene: PDE8A | 212522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1434
p=0.2613
p=0.0011
p=1e−04
p=0.0486
p=0.0377
p=2e−04
p=0.3196
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

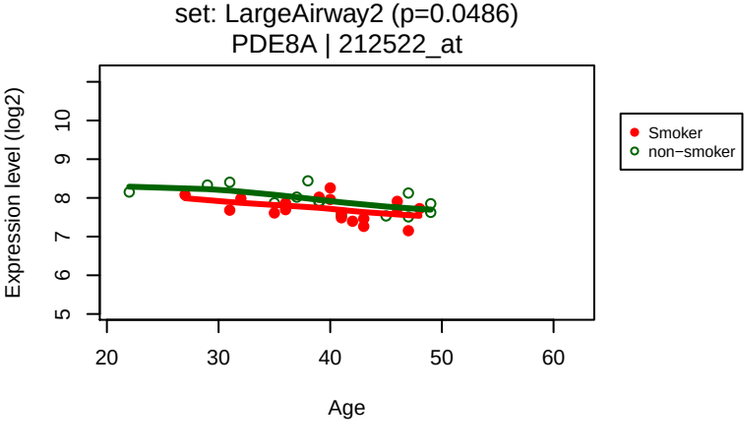

Gene expression ratio (smokers/nonsmokers)
gene: PDE8A | 212522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1434
p=0.2613
p=0.0011
p=1e−04
p=0.0486
p=0.0377
p=2e−04
p=0.3196
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

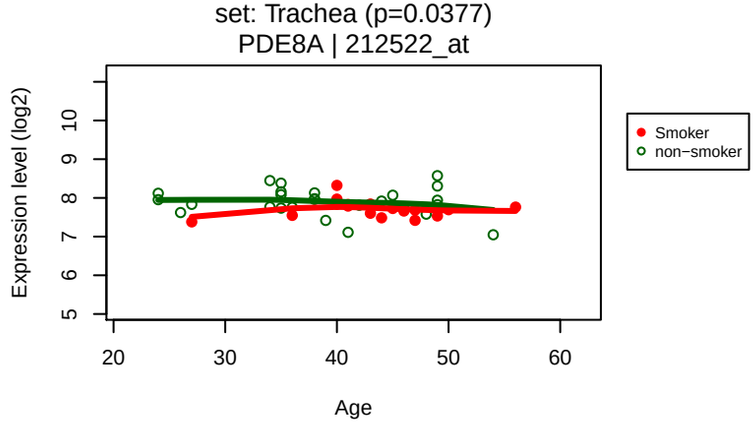

Gene expression ratio in Males
gene: PDE8A | 212522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7346
p=0.0494
p=5e−04
p=0.5319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE8A | 212522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3752
p=0.8264
p=0.4616
p=0.4997
SmallAirway2
LargeAirway2
Trachea
Macrophage2

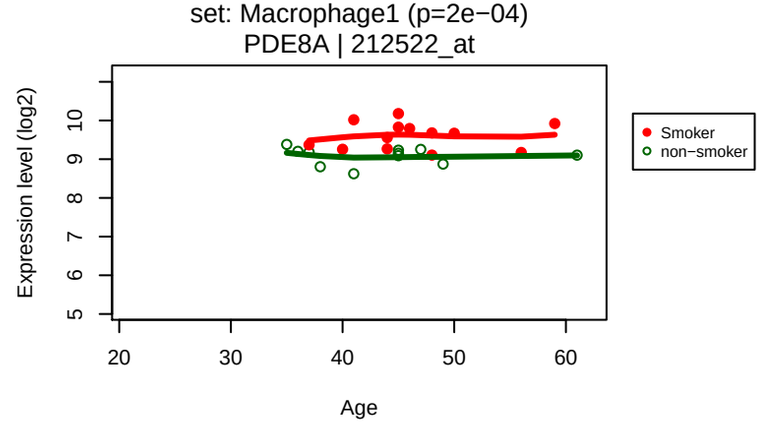

set: SmallAirway1 (p=0.1434)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker

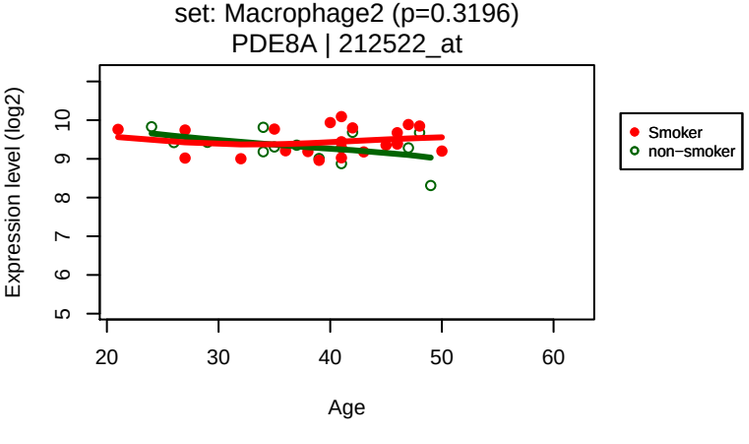

set: SmallAirway2 (p=0.2613)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker

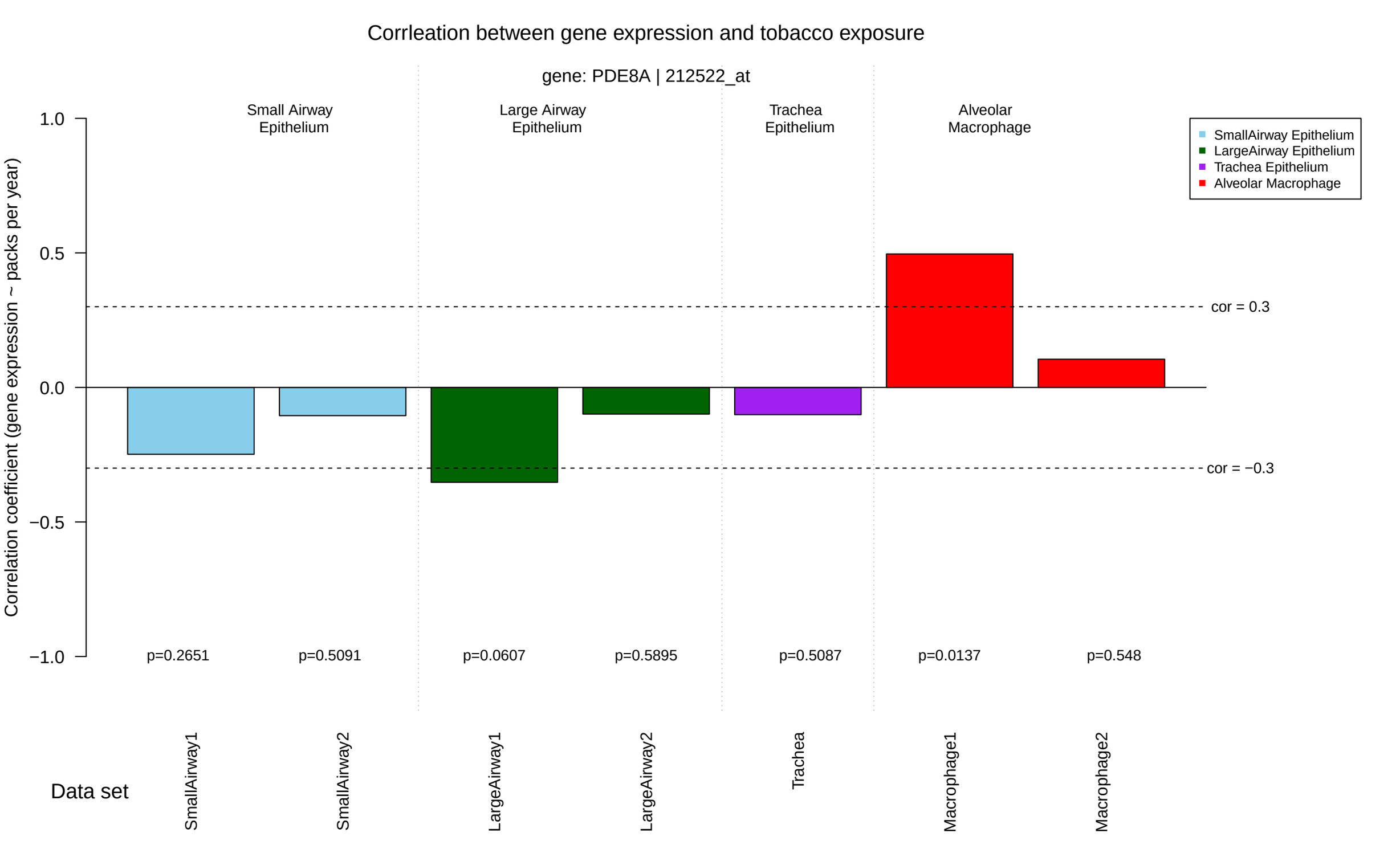

set: LargeAirway1 (p=1e−04)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0486)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0377)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=2e−04)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3196)
PDE8A | 212522_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE8A | 212522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2651
p=0.5091
p=0.0607
p=0.5895
p=0.5087
p=0.0137
p=0.548
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2